

INFORMATION SYSTEMS CASE STUDY



SuratJalan Digital

A Role-Based Delivery Note Management System for Baropi Group

From paper-based delivery notes to a digital, role-based, multi-business-unit workflow.

Executive Summary

Baropi Group is a multi-brand F&B and hospitality company with several internal divisions and six outlets, relying on a fully paper-based delivery note process for internal and external goods movement. This project covers the end-to-end design of a digital replacement — from requirements analysis to a working prototype.

My Role

- Stakeholder interviews & requirements elicitation
- Business process mapping (as-is vs. to-be)
- Conceptual data modeling & access-control design
- Functional specification & low-code prototype build

10+

Divisions & outlets onboarded

2

Signature checkpoints per document

0

Paper copies required after go-live

Problem Statement



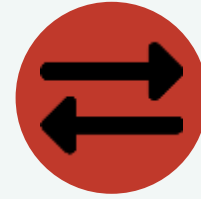
Paper Waste

Every shipment required a duplicate 2-copy carbon form (white & pink).



Poor Traceability

Finding an old delivery note meant digging through physical archives.



Slow Confirmation

Sender had no visibility into delivery status without a phone call.



No Access Control

Physical forms could be signed by anyone present — no real authorization check.

Project Objectives



Digitize the Workflow

Replace physical forms and wet-ink signatures with a fully digital, mobile-friendly workflow.



Enforce Real Authorization

Ensure only the officially registered PIC of a division/outlet can act or sign on its behalf.



Improve Visibility

Give every party real-time status of goods in transit without manual follow-up.



Preserve Audit History

Keep a searchable, tamper-evident record of every transaction, exportable to PDF on demand.

Stakeholder Overview



Internal Divisions

- Marketing
- HR
- Warehouse
- CK



Outlets / Brands

- Shabu Pot
- Baropi Grill (PS/PTC/Opi)
- Barong Resto
- Barong Hotel



External Parties

- Printing vendors
- Ingredient suppliers
- Other 3rd parties

*No system account — access designed separately
(see slide 11)*

As-Is vs. To-Be Process

AS-IS — Paper-Based

- ✗ Staff manually fills out a 2-copy carbon delivery note
- ✗ Sender signs, courier physically carries goods + form
- ✗ Receiver signs on arrival — no real-time confirmation
- ✗ Copies filed manually; retrieval requires physical search
- ✗ No enforced check on who is authorized to sign

TO-BE — Digital System

- ✓ Sender's account is locked to their own division/outlet
- ✓ Digital signature captured directly on-screen
- ✓ Status auto-updates to "In Transit", receiver notified instantly
- ✓ Full text-searchable history, filterable by status/date
- ✓ Signature restricted to the registered PIC of each party

Functional Requirements



Delivery Note Management

- Create, save as draft, or submit directly
- Dynamic item table with unit-aware quantity validation (whole numbers for pcs, decimals for kg/L)



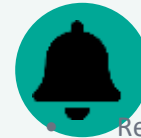
Digital Signature & Workflow

- On-screen signature capture for Sender & Receiver
- Sequential status flow: Draft → Sent → In Transit → Received → Completed



Access & Authorization

- Sender locked to logged-in user's division/outlet
- Signature restricted to the registered PIC of each party



Notification & Reporting

- Real-time in-app alerts on status change
- Filterable dashboard, scoped per account; PDF export on completion

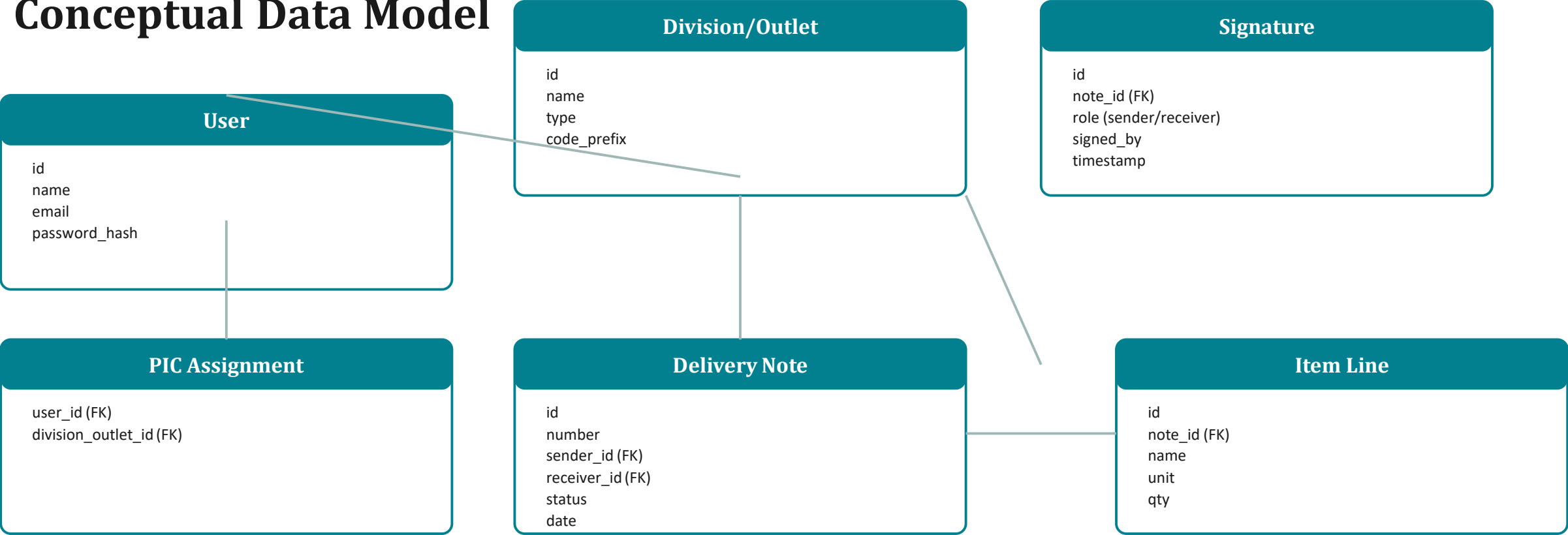
Actors & Role-Based Access Control (RBAC)

Role	Can Create Note As	Can Sign As	Dashboard Scope
Division/Outlet PIC	Own division/outlet only	Own division/outlet (sender or receiver side)	Notes involving own division/outlet
Staff (non-PIC)	Not permitted	Not permitted	Read-only, own division/outlet
External (shared account)	N/A — assigned per transaction	"External" party, on a handed-over device	N/A
Super Admin	Any division/outlet (edge cases)	Not applicable	Full cross-organizational visibility



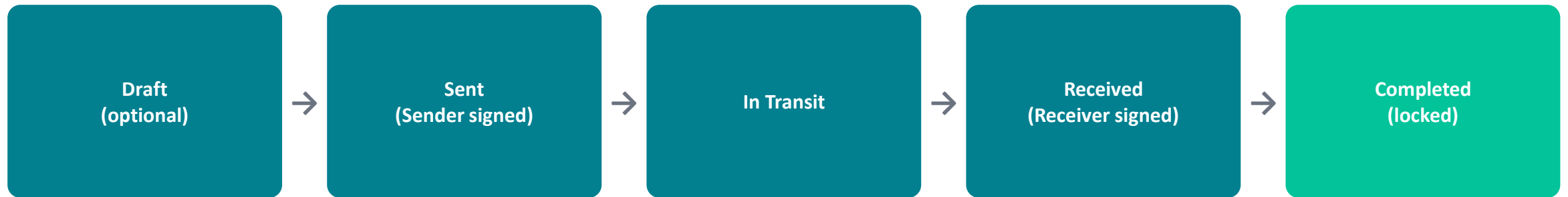
Design principle: authorization is checked against the specific sender/receiver of each document, not just the logged-in identity — a user with no relation to a note cannot act on it, regardless of role.

Conceptual Data Model



Simplified — full schema also includes Unit reference table and Attachment (photos).

Document Status Lifecycle



Key constraint: the Receiver's signature field is disabled by the system until the Sender has signed and clicked "Send" — enforcing correct sequencing in software, not just in policy.

Handling External (Non-Employee) Parties

Two design options were evaluated for parties without system accounts (vendors, suppliers):

Option A — Public Link + QR Token

- Requires token generation, expiry logic, and a public unauthenticated page
- Higher build & maintenance cost
- Not selected — added complexity outweighed benefit at current scale

Option B — Shared "External" Account (Selected)

- ✓ One shared account authorized as "External" for any non-employee transaction
- ✓ Staff hands over a device already logged in; the external party signs directly
- ✓ Mirrors the trust model of the original paper hand-off — no new infrastructure needed

Tools & Approach



Requirements & Analysis

Stakeholder interviews, process mapping, functional specification writing



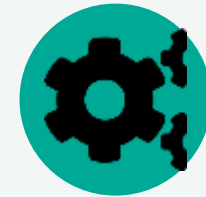
System Design

Conceptual data modeling, RBAC matrix, state-flow design



Rapid Prototyping

Low-code platform (Lovable) for working UI, database, and auth logic



Iterative Refinement

Multiple revision cycles based on stakeholder feedback before rollout

Prototype: digitalbaropi.lovable.app

Outcomes & Key Takeaways

Impact

- ✓ Eliminated duplicate paper forms across 10+ divisions/outlets
- ✓ Real-time delivery status, replacing manual phone follow-ups
- ✓ Fully searchable, exportable digital audit trail
- ✓ Access restricted to verified, registered signatories only

Skills Demonstrated

- ✓ Requirements elicitation & stakeholder communication
- ✓ Business process (as-is/to-be) modeling
- ✓ Conceptual data modeling & RBAC design
- ✓ Trade-off evaluation between design alternatives
- ✓ Rapid prototyping & iterative delivery

Next Steps: production hosting on a custom domain, and extending the shared-external-account model with per-vendor audit tagging.